

```
#starter code
from pyspark.sql import SparkSession

spark = SparkSession.builder \
    .appName("Phase2") \
    .getOrCreate()

customers = spark.read.option("header", "true").csv("/samples/customers.csv")
sales = spark.read.option("header", "true").csv("/samples/sales.csv")

customers.show()
sales.show()
customers.printSchema()
sales.printSchema()
customers = customers.dropna(subset=["customer_id"])
customers.show()
sales = sales.dropna(subset=["customer_id"])
sales.show()
```

```
+-----+-----+-----+-----+-----+-----+-----+-----+
|customer_id|first_name|last_name|email|phone_number|address|city|state|zip_code|
+-----+-----+-----+-----+-----+-----+-----+-----+
| 1| John| Smith|john.smith@domain...| 555-0001| 123 Elm St| Springfield| IL| 62701|
| 2| Emma| Jones|emma.jones@webmai...| 555-0002| 456 Oak St| Centerville| OH| 45459|
| 3| Olivia| Brown|olivia.brown@outl...| 555-0003| 789 Pine St| Greenville| SC| 29601|
| 4| Liam| Johnson|liam.johnson@gmai...| 555-0004| 101 Maple St| Riverside| CA| 92501|
| 5| Noah| Williams|noah.williams@yah...| 555-0005| 202 Birch St| Lakeside| TX| 75001|
| 6| Alice| Miller|alice.miller@aol.com| 555-0006| 303 Cedar St| Oakland| CA| 94601|
| 7| Isabella| Davis|isabella.davis@ic...| 555-0007| 404 Spruce St| Boise| ID| 83701|
| 8| James| Martinez|james.martinez@li...| 555-0008| 505 Walnut St| Des Moines| IA| 50301|
| 9| Sophia| Garcia|sophia.garcia@zoh...| 555-0009| 606 Cherry St| Albany| NY| 12201|
| 10| Lucas| Rodriguez|lucas.rodriguez@h...| 555-0010| 707 Maple St| Portland| OR| 97201|
| 11| Mia| Lopez| mia.lopez@mail.com| 555-0011| 808 Oak St| Miami| FL| 33101|
| 12| William| Anderson|william.anderson@...| 555-0012| 909 Pine St| Nashville| TN| 37201|
| 13| Amelia| Thomas|amelia.thomas@pro...| 555-0013| 1010 Elm St| Denver| CO| 80201|
| 14| Ethan| Taylor|ethan.taylor@inbo...| 555-0014| 1111 Birch St| Minneapolis| MN| 55401|
| 15| Harper| Jackson|harper.jackson@ou...| 555-0015| 1212 Cedar St| Seattle| WA| 98101|
| 16| Jackson| White|jackson.white@yma...| 555-0016| 1313 Spruce St| Atlanta| GA| 30301|
| 17| Charlotte| Harris|charlotte.harris@...| 555-0017| 1414 Walnut St| San Diego| CA| 92101|
| 18| Oliver| Martin|oliver.martin@icl...| 555-0018| 1515 Cherry St| Indianapolis| IN| 46201|
| 19| Madison| Thompson|madison.thompson@...| 555-0019| 1616 Maple St| Charlotte| NC| 28201|
| 20| Elijah| Garcia|elijah.garcia@zoh...| 555-0020| 1717 Oak St| Detroit| MI| 48201|
+-----+-----+-----+-----+-----+-----+-----+-----+
only showing top 20 rows
```

only showing top 20 rows

sale_id	customer_id	product_id	sale_date	quantity	total_amount
1	1	1	2024-01-15	2	39.98
2	1	3	2024-01-20	1	29.99
3	2	2	2024-01-16	1	25.00
4	2	4	2024-01-22	3	89.97
5	3	5	2024-01-17	2	49.98
6	4	6	2024-01-18	4	119.96
7	4	7	2024-01-25	1	15.50
8	5	8	2024-01-19	3	66.75
9	6	9	2024-01-20	2	40.00
10	7	10	2024-01-21	5	110.95
11	8	11	2024-01-22	1	20.00
12	9	12	2024-01-23	4	79.96
13	10	13	2024-01-24	2	55.00
14	11	14	2024-01-25	1	25.00
15	12	15	2024-01-26	3	67.47
16	13	16	2024-01-27	2	34.00
17	14	17	2024-01-28	1	15.00
18	15	18	2024-01-29	4	92.00
19	16	19	2024-01-30	3	60.00
20	17	20	2024-01-31	2	40.00

only showing top 20 rows

```
root
|-- customer_id: string (nullable = true)
|-- first_name: string (nullable = true)
|-- last_name: string (nullable = true)
|-- email: string (nullable = true)
|-- phone_number: string (nullable = true)
|-- address: string (nullable = true)
|-- city: string (nullable = true)
|-- state: string (nullable = true)
|-- zip_code: string (nullable = true)
```

```
root
|-- sale_id: string (nullable = true)
|-- customer_id: string (nullable = true)
|-- product_id: string (nullable = true)
|-- sale_date: string (nullable = true)
|-- quantity: string (nullable = true)
|-- total_amount: string (nullable = true)
```

```

+-----+-----+-----+-----+-----+-----+-----+
|customer_id|first_name|last_name|email|phone_number|address|city|state|zip_code|
+-----+-----+-----+-----+-----+-----+-----+
|1|John|Smith|john.smith@domain...|555-0001|123 Elm St|Springfield|IL|62701|
|2|Emma|Jones|emma.jones@webmai...|555-0002|456 Oak St|Centerville|OH|45459|
|3|Olivia|Brown|olivia.brown@outl...|555-0003|789 Pine St|Greenville|SC|29601|
|4|Liam|Johnson|liam.johnson@gmai...|555-0004|101 Maple St|Riverside|CA|92501|
|5|Noah|Williams|noah.williams@yah...|555-0005|202 Birch St|Lakeside|TX|75001|
|6|Alice|Miller|alice.miller@aol.com|555-0006|303 Cedar St|Oakland|CA|94601|
|7|Isabella|Davis|isabella.davis@ic...|555-0007|404 Spruce St|Boise|ID|83701|
|8|James|Martinez|james.martinez@li...|555-0008|505 Walnut St|Des Moines|IA|50301|
|9|Sophia|Garcia|sophia.garcia@zoh...|555-0009|606 Cherry St|Albany|NY|12201|
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|11|Mia|Lopez|mia.lopez@mail.com|555-0011|808 Oak St|Miami|FL|33101|
|12|William|Anderson|william.anderson@...|555-0012|909 Pine St|Nashville|TN|37201|
|13|Amelia|Thomas|amelia.thomas@pro...|555-0013|1010 Elm St|Denver|CO|80201|
|14|Ethan|Taylor|ethan.taylor@inbo...|555-0014|1111 Birch St|Minneapolis|MN|55401|
|15|Harper|Jackson|harper.jackson@ou...|555-0015|1212 Cedar St|Seattle|WA|98101|
|16|Jackson|White|jackson.white@yma...|555-0016|1313 Spruce St|Atlanta|GA|30301|
|17|Charlotte|Harris|charlotte.harris@...|555-0017|1414 Walnut St|San Diego|CA|92101|
|18|Oliver|Martin|oliver.martin@icl...|555-0018|1515 Cherry St|Indianapolis|IN|46201|
|19|Madison|Thompson|madison.thompson@...|555-0019|1616 Maple St|Charlotte|NC|28201|
|20|Elijah|Garcia|elijah.garcia@zoh...|555-0020|1717 Oak St|Detroit|MI|48201|
+-----+-----+-----+-----+-----+-----+-----+
only showing top 20 rows

```

```

+-----+-----+-----+-----+-----+-----+
|sale_id|customer_id|product_id|sale_date|quantity|total_amount|
+-----+-----+-----+-----+-----+-----+
|1|1|1|2024-01-15|2|39.98|
|2|2|3|2024-01-20|1|29.99|
|3|3|2|2024-01-16|1|25.00|
|4|4|4|2024-01-22|3|89.97|
|5|5|3|2024-01-17|2|49.98|
|6|6|4|2024-01-18|4|119.96|
|7|7|4|2024-01-25|1|15.50|
|8|8|5|2024-01-19|3|66.75|
|9|9|6|2024-01-20|2|40.00|
|10|10|7|2024-01-21|5|110.95|
|11|11|8|2024-01-22|1|20.00|
|12|12|9|2024-01-23|4|79.96|
|13|13|10|2024-01-24|2|55.00|
|14|14|11|2024-01-25|1|25.00|
|15|15|12|2024-01-26|3|67.47|
|16|16|13|2024-01-27|2|34.00|
|17|17|14|2024-01-28|1|15.00|
|18|18|15|2024-01-29|4|92.00|
|19|19|16|2024-01-30|3|60.00|
|20|20|17|2024-01-31|2|40.00|
+-----+-----+-----+-----+-----+-----+
only showing top 20 rows

```

```
1 #starter code
2 from pyspark.sql import SparkSession
3
4 spark = SparkSession.builder \
5     .appName("Phase2") \
6     .getOrCreate()
7
8 customers = spark.read.option("header", "true").csv("/samples/customers.csv")
9 sales = spark.read.option("header", "true").csv("/samples/sales.csv")
10
11
12 #1)Total order amount for each customer
13 from pyspark.sql.functions import sum
14
15 total_orders_df = sales.groupBy("customer_id") \
16     .agg(sum("total_amount").alias("total_order_amount"))
17
18 total_orders_df.show()
```

```

+-----+-----+
|customer_id|total_order_amount|
+-----+-----+
|          7|          110.95|
|         15|           92.0|
|         11|           25.0|
|         29|          35.98|
|         42|           22.5|
|          3|          49.98|
|         30|          71.97|
|         34|           92.0|
|          8|           20.0|
|         22|         119.96|
|         28|           45.0|
|         16|           60.0|
|         35|           60.0|
|         43|           40.0|
|          5|          66.75|
|         31|           50.0|
|         18|          59.97|
|         27|          79.96|
|         17|           40.0|
|         26|          66.75|
+-----+-----+

```

only showing top 20 rows

```

11
12 #Top 3 customers by total spend
13 from pyspark.sql.functions import sum
14
15 top_3 = sales.groupBy("customer_id") \
16     .agg(sum("total_amount").alias("total_spend")) \
17     .orderBy("total_spend", ascending=False) \
18     .limit(3)
19
20 top_3.show()
21
22 |

```

```

+-----+-----+
|customer_id|total_spend|
+-----+-----+
|          4|135.45999999999998|
|          22|119.96|
|          2|114.97|
+-----+-----+

```

```

#starter code
from pyspark.sql import SparkSession

spark = SparkSession.builder \
    .appName("Phase2") \
    .getOrCreate()

customers = spark.read.option("header", "true").csv("/samples/customers.csv")
sales = spark.read.option("header", "true").csv("/samples/sales.csv")

#3. Customers with no orders
no_orders = customers.join(
    sales,
    on="customer_id",
    how="left"
).filter(sales.customer_id.isNull())

no_orders.show()

```

customer_id	first_name	last_name	email	phone_number	address	city	state	zip_code	sale_id	product_id	sale_date	quantity	total_amount
47	Amelia	James	amelia.james@live...	555-0047	4444 Oak St	San Antonio	TX	78202	NULL	NULL	NULL	NULL	NULL
48	James	Walker	james.walker@mai...	555-0048	4545 Pine St	Dallas	TX	75201	NULL	NULL	NULL	NULL	NULL
49	Elena	Gray	elena.gray@zoho.com	555-0049	4646 Elm St	Detroit	MI	48202	NULL	NULL	NULL	NULL	NULL
50	Henry	Phillips	henry.phillips@ao...	555-0050	4747 Birch St	Indianapolis	IN	46202	NULL	NULL	NULL	NULL	NULL

```
1 #starter code
2 from pyspark.sql import SparkSession
3
4 spark = SparkSession.builder \
5     .appName("Phase2") \
6     .getOrCreate()
7
8 customers = spark.read.option("header", "true").csv("/samples/customers.csv")
9 sales = spark.read.option("header", "true").csv("/samples/sales.csv")
10
11
12 #4. City-wise total revenue
13 from pyspark.sql.functions import sum
14
15 # 4. City-wise total revenue
16 city_revenue = sales.join(customers, on="customer_id", how="inner") \
17     .groupBy("city") \
18     .agg(sum("total_amount").alias("total_city_revenue"))
19
20 city_revenue.show()
```



```
+-----+-----+
|      city|total_city_revenue|
+-----+-----+
|  Springfield|      69.97|
|    Oakland|      40.0|
| San Antonio|      79.96|
| Philadelphia|      60.0|
| Indianapolis|      59.97|
|   Nashville|      67.47|
|   San Diego|      95.5|
|San Francisco|      27.5|
|    Detroit|      89.97|
|    Albany|      79.96|
|   Portland|      55.0|
|    Boise|     110.95|
|   St. Louis|      99.95|
|   Columbus|      71.97|
|   Memphis|      89.96|
| Washington|      92.0|
|   El Paso|      19.99|
| Milwaukee|      66.75|
| Pittsburgh|      15.5|
| Centerville|     114.97|
+-----+-----+
```

only showing top 20 rows

```
1 #starter code
2 from pyspark.sql import SparkSession
3
4 spark = SparkSession.builder \
5     .appName("Phase2") \
6     .getOrCreate()
7
8 customers = spark.read.option("header", "true").csv("/samples/customers.csv")
9 sales = spark.read.option("header", "true").csv("/samples/sales.csv")
10
11
12 # . Average order amount per customer
13 from pyspark.sql.functions import avg
14 avg_amount=sales.groupBy("customer_id")\
15     .agg(avg("total_amount").alias("average_amount"))
16 avg_amount.show()
```

```

+-----+-----+
|customer_id|average_amount|
+-----+-----+
|          7|         110.95|
|         15|          92.0|
|         11|          25.0|
|         29|         35.98|
|         42|          22.5|
|          3|         49.98|
|         30|         71.97|
|         34|          92.0|
|          8|          20.0|
|         22|        119.96|
|         28|          45.0|
|         16|          60.0|
|         35|          60.0|
|         43|          40.0|
|          5|         66.75|
|         31|          50.0|
|         18|        29.985|
|         27|         79.96|
|         17|          40.0|
|         26|         66.75|

```

```

+-----+-----+

```

only showing top 20 rows

#6. Customers with more than one order

```
from pyspark.sql.functions import count, col

cust_morethan_one_order = sales.groupBy("customer_id") \
    .agg(count("customer_id").alias("no_of_orders")) \
    .filter(col("no_of_orders") >= 2)

cust_morethan_one_order.show()
```

customer_id	no_of_orders
18	2
1	2
4	2
2	2

Write, Run, and Test Code Effectively

```
1  #starter code
2  from pyspark.sql import SparkSession
3
4  spark = SparkSession.builder \
5      .appName("Phase2") \
6      .getOrCreate()
7
8  customers = spark.read.option("header", "true").csv("/samples/customers.csv")
9  sales = spark.read.option("header", "true").csv("/samples/sales.csv")
10
11
12  #7. Sort customers by total spend descending
13  from pyspark.sql.functions import sum
14
15  top_3 = sales.groupBy("customer_id") \
16      .agg(sum("total_amount").alias("total_spend")) \
17      .orderBy("total_spend", ascending=False)
18
19
20  top_3.show()
```

```
+-----+-----+
|customer_id|    total_spend|
+-----+-----+
|         4|135.45999999999998|
|        22|         119.96|
|         2|         114.97|
|         7|         110.95|
|        45|         109.96|
|        24|          99.95|
|        15|          92.0|
|        34|          92.0|
|        20|          89.97|
|        39|          89.96|
|         9|          79.96|
|        27|          79.96|
|        30|          71.97|
|         1|          69.97|
|        12|          67.47|
|        41|          67.47|
|         5|          66.75|
|        26|          66.75|
|        35|          60.0|
|        16|          60.0|
+-----+-----+
only showing top 20 rows
```